

Create Process Escape Sequences, Handle Escaped Spaces, Preserve Characters, Validate Parser Output, Test Complex Inputs

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    char output[100];
    int i = 0, j = 0;

    printf("Enter text: ");
    fgets(input, sizeof(input), stdin);

    while (input[i] != '\0') {
        if (input[i] == '\\') {
            i++;

            if (input[i] == ' ' ||
                input[i] == '\\\ ' ||
                input[i] == '"' ||
                input[i] == '\') {
                output[j++] = input[i];
            }
            else {
                printf("Invalid escape sequence\n");
                return 1;
            }
        }
    }
}
```

```

        }
    }
    else {
        output[j++] = input[i];
    }

    i++;
}

output[j] = '\0';

printf("Parsed text: %s", output);
printf("Parsing successful\n");

return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill/Week 3$ ./task3
Enter text: hello\\e
Parsed text: hello\e
Parsing successful
geya@LenovoL480:~/OS/Skill/Week 3$ ./task3
Enter text: hello\\\d
Invalid escape sequence
geya@LenovoL480:~/OS/Skill/Week 3$ |

```